



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2106-1- SIGNALS AND SYSTEMS

(For ACT, ECE, VDT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

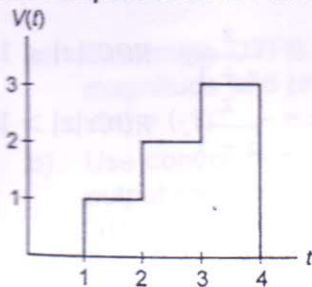
1. The first derivative of a unit step function is _____

- | | |
|--------------------------|-----------------------|
| A) Unit Impulse function | B) Unit Step Function |
| C) Rectangular Function | D) Sinc Function |

2. Convolution of $x(n) * \delta(n) =$

- | | |
|----------------|-----------|
| A) $u(n)$ | B) $s(n)$ |
| C) $\delta(n)$ | D) $x(n)$ |

3. The expression for the given waveform is



- | | |
|-----------------------------------|-------------------------------------|
| A) $u(t-1)+u(t-2)+u(t-3)$ | B) $u(t-1)+2u(t-2)+3u(t-3)-3u(t-4)$ |
| C) $u(t-1)+u(t-2)+u(t-4)-3u(t-4)$ | D) $u(t-1)+u(t-2)+u(t-3)-3u(t-4)$ |

4. $x(t) \cdot \delta(t)$ is = _____

- | | |
|---------------------------|---------------------------|
| A) $x(0) \cdot \delta(t)$ | B) $x(t) \cdot \delta(t)$ |
| C) $x(t) \cdot \delta(0)$ | D) 0 |

5. The distributive property of convolution sum is given by

- | | |
|---|---|
| A) $x(n) * \{h_1(n) + h_2(n)\} = x(n) * h_1(n) + x(n) * h_2(n)$ | B) $x(n) * \{h_1(n) * h_2(n)\} = x(n) * h_1(n) + x(n) * h_2(n)$ |
| C) $x(n) * \{h_1(n) * h_2(n)\} = \{x(n) * h_1(n)\} * h_2(n)$ | D) None of these |

6. A discrete time LTI system with impulse response $h[n] = \sin\left(\frac{\pi}{2}n\right)$ is

- | | |
|------------------------------------|---------------------------------|
| A) Memory, causal & stable | B) Has memory, causal, unstable |
| C) Has memory, noncausal, unstable | D) Memoryless, causal, stable |

7. The linearity property of Discrete time Fourier Series is $ax(n)+by(n)=$

- | | |
|--------------------------|--------------------------|
| A) $aX(K)+bY(K)$ | B) $aX(K-K_0)+bY(K)$ |
| C) $aX(K-K_0)+bY(K-K_0)$ | D) $aX(K+K_0)+bY(K+K_0)$ |

8. The fundamental period of $x(t) = \sin^2 t$ is. _____

- | | |
|------------|-----------|
| A) π | B) 2π |
| C) $\pi/2$ | D) 2 |

9. What are the properties of continuous time Fourier series?

- | | |
|--|---|
| A) Linearity, time shifting | B) Linearity, time shifting, frequency shifting |
| C) Linearity, time shifting, frequency shifting, time reversal, time scaling, periodic convolution | D) Linearity, time shifting, frequency shifting, time reversal, time scaling, periodic convolution, multiplication, differentiation |

10. What is the outcome of a periodic convolution of signals in case of continuous time Fourier series?
 A) Division in Frequency domain B) Multiplication in the time domain
 C) Multiplication in the time domain by a sinusoid D) Multiplication in frequency domain
11. Parseval's theorem with reference to DTFT describes
 A) Power of a signal B) Energy of a signal
 C) Length of the signal D) None of these
12. If $x[n]$ is a signal whose DTFT is $X(\Omega)$, then if $y[n] = e^{j\Omega_0 n}$ is a signal, its DTFT is
 A) $X(\Omega - K_0)$ B) $X(n - n_0)$
 C) $X(k - \Omega_0)$ D) $X(\Omega - \Omega_0)$
13. Find the Fourier transform of an exponential signal $f(t) = e^{-at} u(t)$, $a > 0$.
 A) $1/a + j\omega$ B) $1/a - j\omega$
 C) $1 - a + j\omega$ D) $1 - a - j\omega$
14. The convolution property of DTFT is given by
 A) $x(n) * y(n) \leftrightarrow X(e^{j\Omega})Y(e^{j\Omega})$ B) $x(n)y(n) \leftrightarrow X(e^{j\Omega}) * Y(e^{j\Omega})$
 C) $x(n)y(n) \leftrightarrow X(e^{j\Omega})Y(e^{j\Omega})$ D) None of these
15. The Duality property DTFS is given by
 A) $X(n) \leftrightarrow \frac{1}{N}x(k)$ B) $X(n) \leftrightarrow \frac{1}{N}x(-k)$
 C) $X(n) \leftrightarrow x(-k)$ D) $X(n) \leftrightarrow \frac{1}{2N}x(-k)$
16. The Z-transform of unit impulse function is
 A) 0 B) -1
 C) 1 D) None of the above
17. If $x[n] = u[n - 2]$, then $X(z)$ is given by
 A) $\frac{z^{-1}}{z - 1}$, $ROC: |z| > 1$ B) $\frac{z}{z + 1}$, $ROC: |z| < 1$
 C) $\frac{z^{-1}}{z - 1}$, $ROC: |z| < 1$ D) $\frac{z}{z - 1}$, $ROC: |z| > 1$
18. The following method is used for finding inverse Z transform
 A) Power transform method B) Runge Kutta method
 C) Partial Fraction Method D) Differential method
19. For a finite length positive time sequence, the ROC is given by
 A) $|z| < 1$ B) $|z| > 0$
 C) $|z| > 1$ D) $|z| < 0$
20. For a negative time infinite length sequence, the pole will be
 A) Exterior to the ROC B) Interior to the ROC
 C) Both (A) & (B) D) Neither (A) nor (B)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Check if the sum of two signals is periodic or non-periodic. Find the fundamental period if periodic.
i. $x(n) = \cos\left(\frac{\pi n}{3}\right) + \cos\left(\frac{\pi n}{4}\right)$
ii. $x(t) = 2 \cos(10\pi t) + 10 \sin(20\pi t)$ | 4 | L*3 | 2 | 1 |
| b) Sketch the following signals.
i. $x(t) = u(t+3) + 2u(t+1) - 2u(t-1) - u(t-3)$
ii. $x(n) = [3 - n] [u(n+2) - u(n-3)]$ | 6 | L2 | 1 | 1 |
| c) Given $x(n) = \{1/2, 1, 1, 2\}$
Perform i) $x(n^2)$, ii) $x^2(n)$, iii) $2x(n)$, iv) $x(2n)$ | 6 | L3 | 1 | 1 |
| 2. a) Find odd & even components of the following signals.
i. $x(n) = \{1, 2, 3, 0, 0, 0, 0, 0, 1\}$
ii. $x(t) = \cos(t) + \sin(t) + \sin(t)\cos(t)$ | 8 | L3 | 1 | 1 |

- b) An LTI system has an impulse response $h(n) = u(n)$ and $x(n) = \{1, 1, 2, 5\}$. Determine output of the given system. 8 L3 2 1
3. a) Evaluate $y(t) = x(t) * h(t)$ where $x(t)$ & $h(t)$ are given below

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 3 \\ 0, & \text{Otherwise} \end{cases}$$

$$h(t) = \begin{cases} 1, & 0 \leq t \leq 2 \\ 0, & \text{Otherwise} \end{cases}$$
 8 L3 2 1
- b) Investigate the causality and stability of the LTI system whose impulse response is given by
 i. $h(n) = (0.5)^n u(n)$
 ii. $h(t) = \cos \pi t u(t)$ 4 L3 2 1
- c) Determine energy or power of the given signal.
 $x(n) = (1/2)^n$ for $n \geq 0$ and $x(n) = 0$ for $n < 0$ 4 L3 1 1

Unit – II

4. a) Evaluate the Fourier series representation of the signal
 $x(t) = 3 \cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$. Calculate the magnitude and phase values and draw the magnitude and phase spectrum. 8 L3 3 1
- b) Determine the Fourier Transform of the signal and sketch the resulting signal
 $x(t) = e^{-at}u(t); a > 0$ 8 L3 4 1
5. a) Determine DTFS co-efficients of the given $x(n)$, sketch $x(n)$ and the magnitude and phase spectrum of the resulting signal
 $x(n) = (-1)^n; -\infty \leq n \leq \infty$ 8 L3 3 1
- b) Use convolution property of Fourier Transform in time domain to find output response $y(t)$ of the system where $h(t) = 3e^{-t}u(t)$ and $x(t) = 2e^{-2t}u(t)$ 8 L3 4 1
6. a) Determine Inverse Discrete Fourier transform of the given signal
 $X(e^{j\Omega}) = 6 / (e^{-j2\Omega} - 5e^{-j\Omega} + 6)$ 8 L3 3 1
- b) Obtain Fourier Transform of the signal $x(t) = \delta(t+0.5) - \delta(t-0.5)$. Sketch $x(t)$ and plot phase and magnitude spectrum of the output signal. 8 L3 4 1

Unit – III

7. a) Find the Z-transform of unit step sequence and plot its Region of Convergence 4 L3 5 1
- b) Find Z-transform of the signal $x(n) = 3(2)^n u(-n)$ using appropriate properties 4 L3 5 1
- c) Compute the inverse Z-transform of $X(z) = \frac{-1+5z^{-1}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}$, ROC: $|z| > 1$. 8 L3 5 1
8. a) Determine the Z-Transform of $x(n) = \left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n)$ and identify the ROC associated with Z-transform of the signal 8 L3 5 1
- b) Find the sequence associated with the following Z-transform using partial fraction expansion, where $x(n)$ is absolutely summable

$$X(z) = \frac{1 - 2z^{-1}}{1 - \frac{5}{2}z^{-1} + z^{-2}}$$

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
